

Direction Forecasting on Daily U.S. Wheat Futures with Machine Learning and Macroeconomic Features

A Comparison of FAVAR, ARX, ELM, and TCN

A preliminary investment strategy and empirical evaluation study under leakage-safe, chronological splits.

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https://github.com/kf-nyu/vip_fall2026 coursework reproduction bundle accompanies the manuscript.

Abstract—We forecast the next-day *direction* of Chicago SRW wheat futures using daily closes, five leakage-safe technicals, and thirty-one FRED-MD macros with stationarity codes and a one-month publication-style lag merged to a daily calendar. Equilibrium forward-curve modeling for storable commodities [1] frames wheat futures as a spot-futures system shaped by storage and inventory, not only a martingale in returns. Four estimators—FAVAR-style factor logistic and VAR-threshold paths, ridge ARX, ELM, and a TCN classifier and regressor—share chronological evaluation: five-fold time-series cross-validation on the pre-holdout slice and a fifteen-percent holdout. The TCN classifier attains the strongest holdout accuracy (≈ 0.54) and macro-averaged F1; fold accuracies near 0.50 temper any claim of a large, stable directional edge under honest splits. *Framing*: we treat this manuscript as an *empirical evaluation study* under leakage-safe, chronological splits: sophisticated nonlinear heads can edge transparent baselines on direction, but the lift is modest, so honest uncertainty summaries and simple exposure rules may matter as much as small accuracy gaps. Throughout, we treat next-day *sign* as a tactical primitive—for example long-flat or hurdle rules keyed to predicted up-probability and simple risk overlays—rather than as a stand-in for full execution or portfolio optimization. We summarize empirical mode decomposition cautions for signed-move targets (Sec. IV), document optional wheat-only and cross-asset PCA-augmented TCN paths in repository code (Sec. V), and report a Bayesian logistic specification (Sec. VI) that supplies posterior class probabilities for threshold-aware positioning alongside the headline estimators, without elevating it to a competing “champion” pipeline. Finally, we include a *preliminary investment strategy* illustration—buy-and-hold versus tactical long-flat rules on terminal holdout *proxy* simple returns (Sec. VIII)—with *no* modeled transaction costs, margin, options, rolls, or other brokerage mechanics; it connects probabilities to positioning in a deliberately small, non-advisory sense only.

Index Terms—Wheat futures, direction forecasting, empirical evaluation study, tactical positioning, leakage-safe design, FRED-MD, FAVAR, ARX, ELM, TCN, time-series cross-validation, macroeconomic factors, Bayesian logistic regression.

I. INTRODUCTION

A. Motivation and scope

Many desks care less about the exact tomorrow price and more about a simple question: will the market be *up* or *down*? That sign is enough for basic rules—stay invested or step aside, maybe only act when a model is confident [3]. The catch is that day-to-day grain prices mostly behave like a coin that is slightly biased at best, so any “edge” is small and easy to fake if the data setup is sloppy.

We study Chicago soft red winter wheat, a heavily watched U.S. benchmark. Commodity futures research often links forward curves and spreads to storage and inventory incentives [1]. The author first studied the CMU GSIA/Tepper working-paper precursor of Routledge, Seppi, and Spatt’s equilibrium forward-curve model in 1999; the manuscript cites the published *Journal of Finance* version for a stable bibliographic anchor. Separate industrial-organization work documents episodes where CBOT/CME wheat delivery terms and convergence to cash strained hedgers [2]. Each day we try to predict whether the next *close-to-close* move is positive, using yesterday’s information: wheat’s own recent path, a handful of technical summaries, and a basket of macro indicators (rates, activity, exchange rates, sentiment, and a few inflation-sensitive series). We keep the goal modest: produce reproducible scores that could feed long-flat or “only trade when the model’s probability of an up day is above a hurdle” ideas. Turning those scores into real dollars (commissions, margin, rolls, etc.) needs extra choices; we park that discussion in Sec. IX.

Primary study vs. exploratory add-ons. The **primary** thread is a head-to-head comparison of four forecasting families on one daily wheat panel with identical chronological discipline (ARX, FAVAR-style heads, ELM, and TCN). **Exploratory** material—still reproducible, but not positioned as a parallel “winner-take-all” race—includes EMD notes for signed targets, optional PCA-augmented TCN code paths (Sec. V), the Bayesian logistic panel for uncertainty and calibration views (Sec. VI), and the terminal long-flat proxy read-out (Sec. VIII).

Contributions. Items (1)–(3) support the **primary** matched study; items (4)–(6) document **exploratory** ex-

tensions and positioning layers. (1) A unified, leakage-aware daily pipeline with 31-variable FRED-MD alignment, t-codes, and a one-month reporting lag. (2) Parallel implementation of four heterogeneous model classes with matched inputs where feasible. (3) Empirical comparison on identical splits, reporting both fold-level cross-validation and a final holdout. (4) Discussion of empirical mode decomposition (EMD) and code-level macro-PCA augmentation paths for TCN (wheat-only versus cross-asset panels; §V). (5) A Bayesian logistic regression (Sec. VI, Appendix G) on ten fold-aligned features exposing uncertainty in $P(y_t=1 \mid \cdot)$, supporting calibrated long-flat or hurdle overlays in the spirit of tactical risk control. It observes the same chronological discipline as the main models but is *not* billed as their substitute “champion” forecaster. (6) A reproducible **buy-and-hold vs. long-flat proxy** read-out on the terminal holdout (Sec. VIII) using $\hat{p}_t$ from `TCN_PCA_WHEAT_ONLY` after train+val-only retraining, *without* margin, rolls, commissions, or other brokerage realism (a follow-on term targets finetuning plus richer execution).

B. Project context

This manuscript doubles as a *VIP Semester Final Report*: it foregrounds pipeline hygiene, honest out-of-sample slicing, and contrasts among interpretable and black-box estimators. Companion project deliverables summarize the project and learning arc; this PDF stays close to methods and metrics. Generative assistants supported code structure, probabilistic-model scripting, and L^AT_EX layout, with all numbers regenerated from repository scripts before submission.

C. Related strands of literature

Classic econometric work on commodity futures emphasizes storage theory, convenience yield, and the term structure of futures prices; reduced-form macro links appear in factor models of commodity co-movement and in vector autoregressions with blocks of financial indicators. Routledge, Seppi, and Spatt [1] model equilibrium forward curves for storable commodities with inventory constraints, giving a reference point for how spot-futures dynamics can depart from simple random-walk benchmarks. Irwin *et al.* [2] discuss episodes of poor futures-cash convergence in CBOT corn, soybean, and *wheat* contracts, a reminder that the Chicago SRW series we download is embedded in exchange rules and delivery economics, not a frictionless continuous price. Machine learning studies often report high *price-level* R^2 using lagged prices alone—a phenomenon frequently driven by persistence rather than genuine multi-step forecasts of returns. Our design separates *direction* scoring from *level* R^2 and stresses honest out-of-sample slices for classification.

Recent temporal convolutional networks provide flexible sequence-to-one mappings for highly structured panels [9];

we adopt TCNs as a strong nonlinear baseline while retaining transparent linear baselines (ARX, regularized logistic FAVAR head) for contrast. Extreme learning machines occupy a middle ground: universal approximation with trainable linear readouts only.

For the exploratory Bayesian logistic slice (Sec. VI), Hamiltonian Monte Carlo and the no-U-turn sampler (NUTS) are standard tools for high-dimensional continuous posteriors [10], [11]; we use them in a small-parameter “dashboard” role rather than as a claim that MCMC itself creates economic alpha.

D. Notation

Let $\mathcal{F}_{t-1}$ denote information available through the end of day $t-1$. All models construct predictors $x_t \in \mathbb{R}^d$ measurable with respect to $\mathcal{F}_{t-1}$ and target the realized next-day move Δ_t or its sign y_t .

II. DATA AND TARGET CONSTRUCTION

A. Prices and technical features

Let P_t denote the settlement or last available daily close. The one-step-ahead *price change* is $\Delta_t = P_t - P_{t-1}$. The *direction label* is $y_t = \mathbf{1}\{\Delta_t > 0\}$. All predictors at index t are constructed using information available strictly before the realization of Δ_t (lagged windows and shifted technicals).

Five technical features mirror the implementation: multi-horizon log return sums, rolling volatility of daily log returns, a smoothed RSI, and a z -score of price relative to a 20-day moving average, each shifted one day to avoid same-bar leakage.

B. FRED-MD macro panel

Thirty-one series were selected to span output, trade, credit, equity valuation, the yield curve and credit spreads, major exchange rates, metals-sensitive producer prices, and consumer sentiment. Each series receives a *t-code* transform (level, difference, log growth, or second difference of logs) following the FRED-MD documentation [4]. Transforms promote approximate stationarity and stabilize linear estimators.

Monthly vintages are merged to the daily wheat calendar by forward fill, then shifted by one additional month to mimic publication delay: a macro observation dated month M is first mapped through t-codes at monthly frequency, then aligned so that daily rows in calendar month M cannot access information that would not yet be available under a simple one-month reporting lag rule implemented in code.

C. Windows, splits, and scaling

For sequence and flattened models except the FAVAR-specific design, each sample uses a lookback of $L=30$ trading days. TCN tensors have shape $(L, 37)$: one channel encoding the lagged close path together with aligned technical and macro channels per time step (implementation

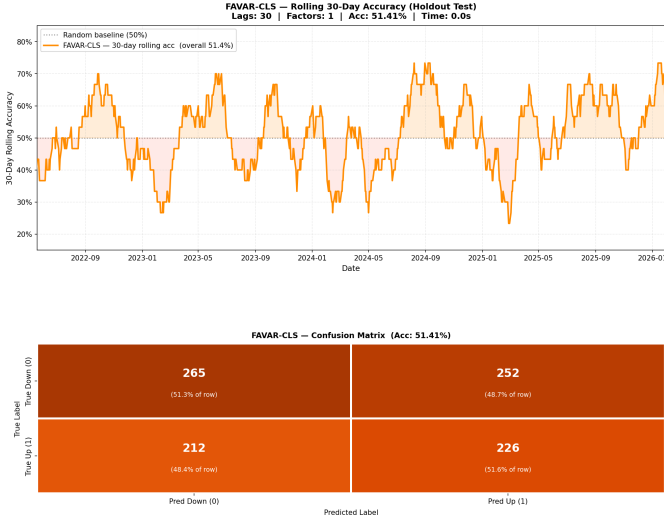


Fig. 1. FAVAR classifier diagnostics ($k=1$ factor, 30 lags).

detail in repository). ARX and ELM flatten to 66 features (30 lags of Δ , five technicals, 31 macro levels at the window end).

The sample is split chronologically: the final 15% of trading days form a **holdout test** (955 observations in our run); the remainder is used for five-fold **TimeSeriesSplit** cross-validation. Feature standardization (and PCA when used) is fit *only* on each training fold and applied to validation and test—never on the full sample prior to split—to prevent leakage of future volatility or location.

III. MODEL FAMILIES

a) Tactical read for Table I.: The classifier columns report $\hat{p}_t = P(y_t=1 | \cdot)$ routed through evaluation-time thresholds tuned inside each training regime; regressors imply similar rules only after $\hat{\Delta}_t$ is converted to $\hat{y}_t$ via inner threshold search. Acc. answers “how often does the chosen rule coincide with realization?”; ROC-AUC summarizes how discriminative $\hat{p}_t$ ranks up versus down days before pinning a hurdle; macro- F_1 penalizes asymmetric errors when one misclassification type is materially worse operationally on a flat-then-long benchmark. Fold dispersion in the CV columns doubles as intuition for sizing: wide ranges suggest cautious capital commitment even when headline holdout accuracy in the same table looks best.

A. FAVAR

The FAVAR path extracts k principal components from the standardized 31-dimensional macro panel (training-only PCA), augments the wheat state, and estimates a linear VAR in extended state for the regression variant; the classification variant applies ℓ_2 -regularized logistic regression on stacked price lags and factors, echoing the spirit of factor-augmented macro vector models [7]. Assumptions

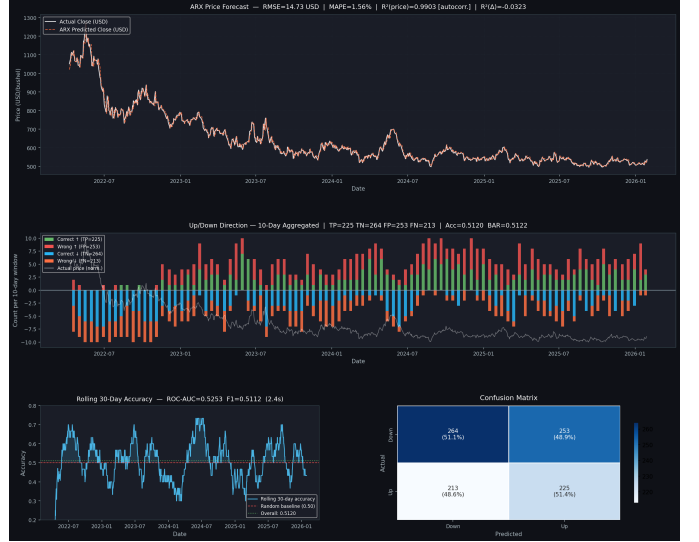


Fig. 2. ARX pipeline visualization (regression and classifier panels).

are linear conditional dynamics and stable Gaussian innovations over long windows—strong in rapidly shifting commodity regimes.

Let $f_t \in \mathbb{R}^k$ denote macro factors extracted as in approximate factor literature [6] and $z_t = [\Delta_t, f_t^\top]^\top$. A VAR(p) writes $z_t = c + \sum_{i=1}^p A_i z_{t-i} + u_t$. One-step forecasts for Δ_t read off the first row of the companion-form impulse dynamics; thresholds convert $\hat{\Delta}_t$ to $\hat{y}_t$. When k is small (we use $k=1$ in the logged run), the system is parsimonious but may omit macro subtleties better captured by the full 31 channels used elsewhere.

Teaching note. The implementation is FAVAR-inspired (macro factors plus lags) rather than a full likelihood-based FAVAR as in Bernanke *et al.* [7]; we retain the label for VIP parallelism with the course vocabulary. Figure 1 (above) plots the $k=1$ classifier diagnostics.

B. ARX

The ARX specification regresses Δ_t on 30 own lags and exogenous macro and technical features with ridge shrinkage, then maps predicted $\hat{\Delta}_t$ to direction using a threshold chosen to maximize macro-F1 on an internal validation slice of each fold. The companion logistic ARX classifier scores $P(y_t=1 | x_t)$ directly.

Ridge mitigates collinearity among overlapping lagged Δ ’s and macro levels; coefficients remain interpretable for teaching purposes (e.g., loadings on term spreads or FX). The main limitation is strict linearity in predictors after fixed transforms.

C. ELM

An extreme learning machine draws random first-layer weights (fixed after initialization), applies a sigmoid non-linearity, and fits a ridge or logistic readout. This yields

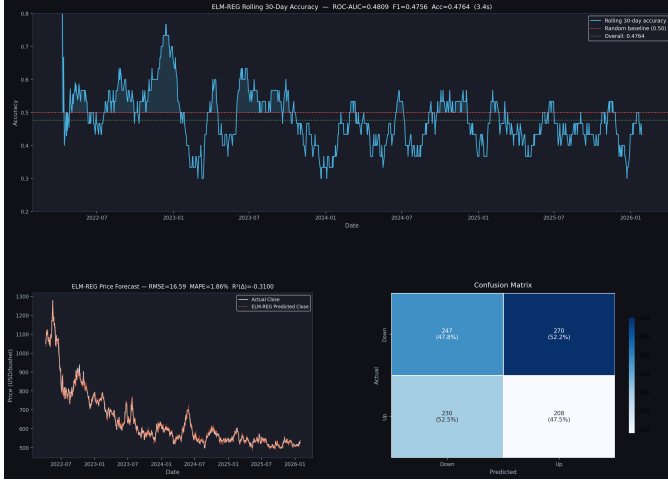


Fig. 3. ELM regressor diagnostics.



Fig. 4. ELM classifier diagnostics.

a fast nonlinear baseline with limited interpretability and sensitivity to hidden width and ridge penalty α .

Let $h_t = \sigma(Wx_t + b)$ with frozen (W, b) and train only output weights β via ridge or logistic loss on h_t [8]. Because W is not gradient-tuned, representation quality is lottery-dependent; multiple seeds would be advisable in production.

D. TCN

A temporal convolutional network stacks dilated causal convolutions over the $L \times 37$ input, followed by a scalar head. The regressor minimizes squared error on Δ ; the classifier minimizes weighted binary cross-entropy with a positive-class weight reflecting mild imbalance between up and down days, and uses a validation-tuned probability threshold at evaluation.

Receptive field grows exponentially in depth; with kernel width three and dilation schedule as implemented, the network can mix information across the entire 30-day window while respecting causality (no future bars inside

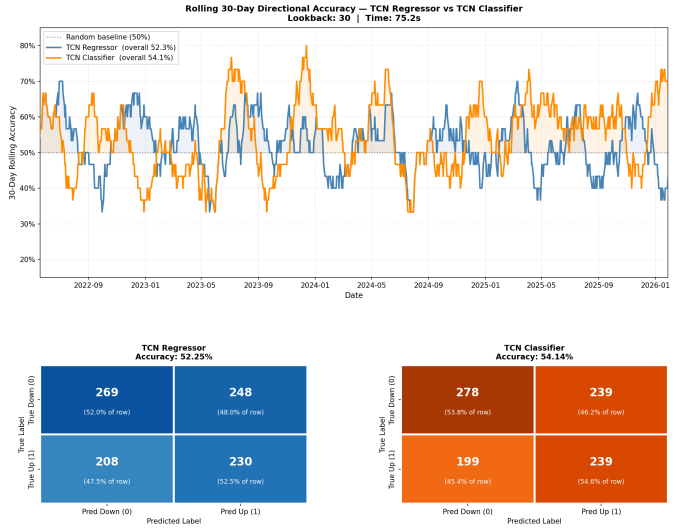


Fig. 5. TCN regressor and classifier outputs for $L=30$.

the window leak into past positions). Dropout and early stopping on validation loss mitigate overfit despite many parameters.

IV. EMD AND DIRECTION

Empirical mode decomposition expresses the price as a sum of intrinsic mode functions (IMFs). Correct aggregation of predicted *signed* moves requires preserving IMF *magnitudes*. Applying independent binary classifiers to each IMF and majority-voting destroys amplitude information and can invert the implied sum [5]. The repository note recommends per-IMF *regression* of increments, summing forecasts, and applying a single terminal threshold—consistent with the observation that EMD helped price-level experiments more than naive IMF-level direction voting.

V. ALTERNATIVE DATA AND PCA FOR TCN

This section documents *what enters principal-component analysis* and what the convolutional head ultimately sees. Both tracks share the leakage discipline of the primary pipeline: **StandardScaler** and (where used) **PCA** are fit only on each fold’s training rows; validation, holdout, and test rows are transformed with those frozen parameters. The supervised target is unchanged: next-bar (or configured-horizon) *sign* of wheat’s log return, as in **TCN_Revised**.

a) *Baseline contrast (TCN_Revised)*.: The baseline tensor is $(L \times 37)$: one wheat **Close** level implicit in differences, five technicals, and thirty-one FRED-MD series after the same t -code and publication-lag rules—*without* Yahoo overlays and *without* fold-wise PCA on macros.

b) *Track (i)—wheat-centered PCA with daily cross-asset returns*.: Script: **TCN/TCN_PCA_WHEAT_ONLY.py**. Seven **wheat price-derived** scalars are built on Chicago

TABLE I

Five-fold time-series CV (mean $\pm$ std of fold validation accuracy) and holdout test (15%, 955 obs.): direction metrics.
Rd: regression $\rightarrow$ threshold; **Clf**: direct classifier.

Model	CV (val.)		Holdout Rd			Holdout Clf		
	Rd	Clf	Acc.	ROC-AUC	F1 _M	Acc.	ROC-AUC	F1 _M
FAVAR	0.495 $\pm$ 0.021	0.502 $\pm$ 0.018	0.513	0.521	0.509	0.514	0.508	0.513
ARX	0.514 $\pm$ 0.018	0.500 $\pm$ 0.016	0.512	0.525	0.511	0.523	0.526	0.522
ELM	0.508 $\pm$ 0.010	0.491 $\pm$ 0.020	0.476	0.481	0.476	0.512	0.528	0.511
TCN	0.504 $\pm$ 0.017	0.499 $\pm$ 0.011	0.523	0.537	0.522	0.541	0.526	0.541

For the best TCN classifier, the holdout confusion matrix was $\begin{bmatrix} 278 & 239 \\ 199 & 239 \end{bmatrix}$, giving slightly higher recall for down days than up days and no strong one-sided failure mode.

TABLE II

Holdout Δ regression quality (USD/day).

Model	RMSE(Δ)	$R^2(\Delta)$
TCN	14.50	-0.0007
ARX	14.73	-0.032
FAVAR	14.47	— [†]
ELM	16.59	-0.310

[†]FAVAR $R^2(\Delta)$ is omitted because the logged screen value was inconsistent with the code definition used for the other rows.

wheat **Close**: **(1–3)** log-returns at 1, 3, and 5 trading days; **(4–5)** rolling standard deviation of the 1-day log-return over 5 and 10 days (realized volatility proxies); **(6–7)** simple momentum $P_t/P_{t-5} - 1$ and $P_t/P_{t-10} - 1$. Four **Yahoo Finance** daily **Close** series are aligned to the same wheat calendar (reindex to wheat dates, forward/backward fill, matching TCN_PCA.py): Chicago corn futures (ZC=F), soybean futures (ZS=F), Invesco USD index ETF (UUP), and CAD/USD spot (CAD=X); each yields a **one-day log-return** feature `corn_ret1`, `soy_ret1`, `uup_ret1`, `cad_ret1`. **Asset choice**. Corn and soy sit in closely related grains/oilseed markets (shared weather, acreage, and export narratives); UUP is a coarse U.S. dollar strength proxy relevant to denominated commodities; CAD=X traces a liquid North-American FX channel (Canada competes with U.S. wheat in export markets). **Functional form**. We use a single short-horizon return per name to avoid bloating PCA inputs (no 3-day or rolling-vol triplet here), reserving that richer Yahoo engineering for track (ii); exploratory same-calendar correlations with wheat were checked with `scripts/wheat_cross_asset_stats.py` (optional `-extended`), supporting these names alongside economic priors rather than formal backward elimination. On each CV fold the *eleven* columns (seven wheat summaries plus the four aligned daily returns) are standardized; PCA (`sklearn.decomposition.PCA`, five components, fixed random seed) fits on training rows only;

the classifier consumes ($L \times 5$) score sequences.

c) *Track (ii)—cross-asset Yahoo + macro PCA.*: Script: TCN/TCN_PCA.py. Per-day features split into three blocks that are **concatenated after** macro reduction:

- **Wheat block (7)**: the same seven wheat-engineered scalars defined at the start of track (i) (log-returns, realized vols, simple momentum). The four cross-asset daily returns are *not* stacked here; they enter only through the macro PCA block below.
- **Yahoo overlay (12)**: WTI crude, COMEX gold, the broad dollar index, and VIX; each expanded to three series—1-day log-return, 3-day log-return, and 5-day rolling volatility of the 1-day log-return (missing columns zeroed after alignment). **Soybean** futures are omitted from this overlay so their signal appears only once, via the macro PCA channel below.
- **Macro PCA block (31+4 $\rightarrow$ 5)**: the same `SELECTED_FEATURES` FRED-MD columns and *t*-code/delay pipeline as `TCN_Revised`, augmented by the **one-day log-returns** of Chicago corn, soybeans, UUP, and CAD=X (`corn_macro_ret1`, `soy_macro_ret1`, `uup_macro_ret1`, `cad_macro_ret1`)—the same economic series and stationarization choice described for track (i). All 35 channels are standardized jointly inside each fold; PCA(5) operates on that stack; five scores feed the TCN head.

Daily rows are therefore $7 + 12 + 5 = 24$ features; reshaped windows are ($L \times 24$). ROC-AUC in Table III is *not* directly interchangeable with the ($L \times 37$) baseline because the information map differs.

Holdout benchmarking for both scripts follows the canonical 85%/15% chronological split when `TCN_HOLDOUT_FRAC=0.15`; Table III summarizes terminal accuracy for the PCA-augmented runs next to baseline `TCN_Revised`. Appendix H records shell replay commands.

VI. EXPLORATORY BAYESIAN LOGISTIC REGRESSION (MCMC)

Tactical overlays—long-flat positioning, hurdle rules on $P(y_t=1)$, and tempered exposure when uncertainty

TABLE III

PCA-augmented TCN on the terminal 15% holdout (train+val-only scalers/PCA; causal inputs before cutoff only). Compare classifier column to Table I. Appendix H lists replay commands.

Pipeline	Acc. [◊]	ROC-AUC	F1 [†] _M
TCN_Revised (reference; Tab. I)	0.541	0.526	0.541
TCN_PCA_WHEAT_ONLY: eleven channels (seven engineered wheat features + corn/soy/UUP/CAD daily log-returns); PCA(5) per fold.	0.543	0.506	0.508
TCN_PCA: Yahoo overlay (12) + macro PCA(5) on FRED-MD + four daily macro returns; 24-d. feature step.	0.482	0.501	0.463

Replay (May 2026): Adam LR=10⁻⁶, 50 epochs, channels {128, 64}, Apple MPS; architecture as Sec. V (corn/soy/UUP/CAD=X daily returns; TCN_PCA now 24-D). Holdout rankings are noisy single tails—TCN_PCA underperforms here versus TCN_PCA_WHEAT_ONLY. ◊Decision threshold 0.5. †Validation-tuned threshold inside each TCNClassifier. ROC-AUC is not directly comparable to Table I because input tensors differ.

is wide versus narrow—consume *class probabilities*, not model architecture summaries. Posterior logistic regression delivers that dispersion on the same chronological splits while exporting full predictive uncertainty for threshold-aware overlays. The MCMC script¹ implements the specification in PyMC v5 using Markov chain Monte Carlo (MCMC) with NUTS [10], [11]: intercept $\alpha \sim \mathcal{N}(0,1)$ and, by default, *global shrinkage* on the p slopes via $\sigma_\beta \sim \text{HalfNormal}(1)$, $\beta_j = \sigma_\beta z_j$ with $z_j \sim \mathcal{N}(0,1)$ (non-centered). Independent Gaussian slabs (BAYES_PRIOR=iid) remain available for low-dimensional designs but produced pathological NUTS geometry when $p = 150$ under `wheat_pca`; the hierarchical prior restores well-behaved sampling (zero divergences, $\hat{R} \approx 1$, bulk ESS in the thousands for default budgets). Calibration and forest panels accompany deterministic forecasters rather than replacing them.

Default design (revised bundle). The MCMC script now defaults to `BAYES_DESIGN=wheat_pca`: the same engineered wheat + cross-asset columns, `create_target` labels, and train+val-only `StandardScaler`+PCA as `TCN_PCA_WHEAT_ONLY`, with each MCMC row flattening the length- L window of PCA scores into $L \cdot K$ features (sequence reshaping is collapsed to a wide vector for the linear logit). This aligns the Bayesian path with the headline PCA-augmented wheat pipeline while remaining a *linear* head rather than the TCN.

Legacy macro design. Setting `BAYES_DESIGN=fred_macro` restores the earlier exploratory block: five scaled short-horizon log returns from the last in-window bar plus five macro PCs from the standardized 31-variable lagged FRED block (PCA fit only on training rows). Table IV and Figures 6–7 use the default `wheat_pca` export ($n_{\text{train}}=5401$, $n_{\text{test}}=958$, 150-D flattened PCA window); regenerate with `BAYES_SAVE_FIG=1`.

The holdout count can differ slightly from the 955 sequence rows in Table I depending on horizon trim-

¹`bayes_logit_mcmc_macro_pca.py`, located in the package root beside README.

TABLE IV

Exploratory Bayesian logit vs. MLE logit (15% holdout; default `wheat_pca`, $n_{\text{test}}=958$, 150-D design).

	Acc.	ROC-AUC	Brier
MLE logit	0.500	0.509	0.255
Bayes (posterior mean p)	0.542	0.508	0.249

ming and window bookkeeping—do not over-interpret one-observation gaps relative to sampling noise.

Posterior *mean* class probabilities are benchmarked against `sklearn`’s `LogisticRegression(C=1)` on the same numeric design. With 2000+2000 tune and draws per chain (two chains, default hierarchical slope prior, `BAYES_TARGET_ACCEPT=0.95`), NUTS reported zero divergences, $\hat{R} \approx 1$, and bulk coefficient ESS in the thousands; longer chains mainly polish Monte Carlo summaries, not test AUC (Table VII).

Comparability caveat. ROC-AUC here (≈ 0.51 , Table IV) lines up with *coin-flip* linear discrimination on this holdout block, unlike the richer TCN slice in Table I. The Bayesian path is for probabilistic overlays on the flattened PCA design, not a claim that the linear head matches sequence-model depth. Holdout **accuracy** at threshold 0.5 can differ sharply between MLE and posterior-mean predictions even when AUC matches, because probabilities need not be calibrated the same way; Brier and ROC curves matter more for ranking.

Graphical diagnostics. Running `bayes_logit_mcmc_macro_pca.py` with the default flag `BAYES_SAVE_FIG=1` regenerates the same PNG diagnostics typeset below as Figures 6–7. Interpretation heuristic for probability-based tactics: ROC curves above the (0,0)–(1,1) diagonal imply $\text{AUC} > 0.5$ discrimination on the held-out timeline; calibration curves hugging that same diagonal mean predicted probabilities align with empirical up-day frequencies bin-by-bin; neither is economic proof yet they signal whether probabilistic summaries are usable for threshold rules. Figures 6–7

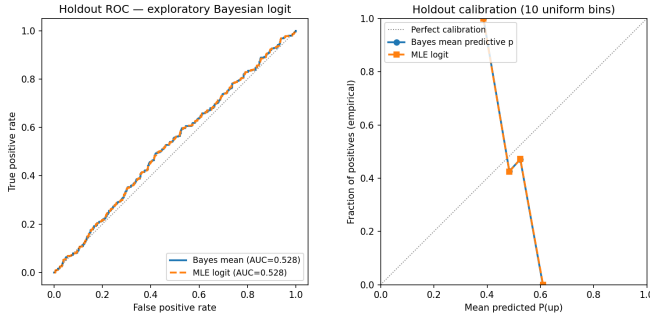


Fig. 6. Bayesian logistic holdout diagnostics (default `wheat_pca`, May 2026; regenerate with `bayes_logit_mcmc_macro_pca.py`). (Left) ROC curves for posterior-mean probabilities vs. MLE logits. (Right) Calibration with ten probability bins spanning $[0, 1]$; deviations diagnose mis-scaled optimism or pessimism for simple long-flat heuristics.

provide those views together with posterior coefficient uncertainty and spread of per-day predictive draws.

Table IV rounds the long-chain posterior-mean run (`BAYES_TUNE = BAYES_DRAWS = 10,000`, two chains) to three decimals; 2000+2000 and 6000+6000 budgets match at this precision (Table VII; `bayes_mcmc_wheat_run_log.md`). A convenient default is `BAYES_DRAWS = BAYES_TUNE = 2000`.

VII. EXPERIMENTAL RESULTS

A. Environment

Final report reproducibility. All tables and figures aggregated into this PDF were produced with `Python 3` from the *project-local virtual environment* (`./venv/`) on **macOS**. TCN training used the Metal Performance Shaders (MPS) device when available.

Earlier course milestones were run in **Google Colab** (CPU); those experiments informed development but are not the canonical source for the numbers frozen here. Minor threshold or loss drifts between Colab CPU and local MPS runs are noted in the validity discussion below.

Hyperparameters follow script defaults: ARX ridge penalty 1.0, ELM hidden width 1000 and $\alpha=0.1$, FAVAR with $k=1$ factor and VAR lag order 3, TCN channels $\{128, 32\}$, kernel size 3, Adam learning rate 3×10^{-4} , early stopping on validation loss.

B. Class imbalance

The pooled training window contained 3038 up days versus 3331 down days (roughly 48% / 52%); metrics emphasize macro-averaged F1 and ROC-AUC alongside accuracy. Directional macro metrics for the four families were reported with Table I in §III; Table II isolates holdout Δ regression error (Tab. II, placed with Tab. I in §III).

a) Threats to validity (expanded):

Information alignment. Although macro series are monthly and shifted, microstructure events (USDA reports, weather model updates) are not encoded in FRED-MD; omitted variables bias coefficients toward zero and can inflate noise in nonlinear heads.

Label noise. Close-to-close returns conflate overnight and session information; calendar effects around USDA release days create heteroskedasticity that i.i.d. loss functions ignore.

Multiple testing. Comparing four families on one holdout slice without multiplicity adjustment risks selective emphasis on the winner; reporting fold distributions partially mitigates but does not eliminate the issue.

Hardware nondeterminism. GPU reductions can introduce bitwise drift across machines; thresholds and early stopping epochs may differ slightly, as observed between Colab CPU and local MPS runs.

Benchmark strength. A seasonal long-short wheat strategy using only calendar dummies or a GARCH(1,1) sign probability might match neural scores; future work should embed such baselines in the same pipeline.

Survivorship and rolls. Continuous front-month stitching can inject artificial jumps at roll dates if not adjusted; our series assumes vendor-provided continuity but users should verify roll rules.

Class threshold dependence. Because direction is derived via scanned thresholds on validation data, reported accuracy is tied to that inner split proportion; sensitivity analysis over 10%–20% internal validation fractions would bolster robustness claims.

C. Qualitative plots

Rolling-accuracy and confusion-matrix exports from the training scripts are typeset in §III immediately after the corresponding model (Figures 1, 2, 3, 4, and 5).

D. Implementation environment

Code paths rely on pure Python stacks—`NumPy/pandas`, `scikit-learn`, `statsmodels` (FAVAR VAR), and `PyTorch` (TCN). The statistics printed in Tables I–II reflect the macOS virtual-environment run described in §VII; prior **Google Colab** sessions used for intermediate homework-style reports reproduced the same qualitative model ordering but occasionally differ by fractions of a percentage point owing to floating-point and hardware backends.

VIII. A PRELIMINARY INVESTMENT STRATEGY—TACTICAL LONG-FLAT OVERLAY (HOLDOUT PROXY P&L)

Section VI argued that *economic* claims require mapping probabilities into positions. We now do so mechanically for the **champion deep classifier** in Table III: `TCN_PCA_WHEAT_ONLY` is retrained on all train+val rows with scaler+PCA fit only on that slice, then evaluated on

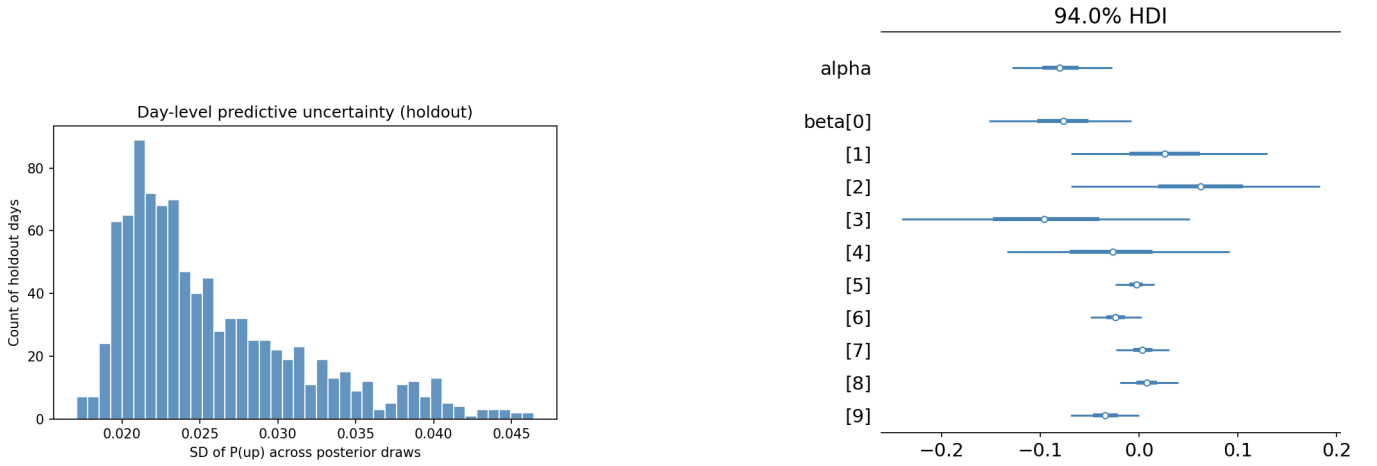


Fig. 7. **(Left)** Holdout dispersion of Bayesian predictive draws; wider histogram mass implies heterogeneous epistemic uncertainty across days—useful stress information before sizing positions using raw probabilities. **(Right)** Coefficient-scale summaries (α , $\beta_j = \sigma_{\beta z_j}$).

the terminal fifteen-percent *sequence* holdout (958 windows in our replay). Let $\hat{p}_t$ denote the TCN’s predicted probability of a strictly positive next-day move (matching `create_target` in code). Positions are $w_t^{\text{BH}} = 1$ (buy-and-hold), $w_t^{(h)} = \mathbf{1}\{\hat{p}_t \geq h\}$ with $h=0.52$ (default hurdle), $w_t^{(0.5)} = \mathbf{1}\{\hat{p}_t \geq 0.5\}$, $w_t^{(\text{thr})} = \mathbf{1}\{\hat{p}_t \geq \tau\}$ where τ is the validation-tuned threshold inside `TCNClassifier`, and a momentum rule $w_t^{\text{mom}} = \mathbf{1}\{y_{t-1}=1\}$ on realized labels. Daily *proxy* returns use the realized simple return on wheat over the same horizon as the label; optional turnover costs are supported in the script but set to zero here. Figure 8 (after the references, for float layout) summarizes Sec. VIII in one schematic: scaler and PCA on train+val, sequence windows into `TCNClassifier`, holdout $\hat{p}_t$, and the long-flat overlay on aligned realized returns. The wheat-PCA Bayesian logit (Sec. VI) shares the same scaler/PCA discipline; `STRATEGY_PANEL=bayes` runs the same hurdle / 0.5 / τ / momentum rules on posterior-mean $\bar{p}_t$ (optional inner-val h^*), while Table V still headlines TCN $\hat{p}_t$ by default. Legacy `BAYES_DESIGN=fred_macro` remains for appendix-style macro designs.

What this simulation omits. This semester’s exercise stops at a *paired comparison*: full exposure (**buy-and-hold**) versus model-timed **long-flat** weights on the same aligned simple returns. We do *not* implement commissions, bid-ask spreads, exchange or clearing fees, contract rolls, margin requirements or margin calls, leverage constraints, listed options, overnight gap risk, position limits, or partial fills—all of which would move any reported P&L closer to a broker-realistic ledger. A follow-on term is expected to pursue stronger model finetuning *and* a richer transaction layer so the strategy read-out approaches executable wheat-futures economics.

Daily cadence, execution scope, and optional Bayes hurdle. The proxy exercise is indexed to *one trading day*

per row after lookback: each holdout day carries one weight w_t and one realized simple return aligned to the same daily bar convention as the codebase (close-to-close style in `create_target` / flat Bayes rows). We do *not* model when during the session prices are observed, when an order could be sent, or whether fills occur at the next open versus the cash close—all of that would be needed for a defensible live timetable. Separately, readers may run `scripts/wheat_strategy_holdout_validation.py` with `STRATEGY_PANEL=bayes` (or `tcn_bayes`) to reuse the *same* long-flat hurdle machinery while swapping the probability input for the *posterior mean* predictive $\bar{p}_t = \mathbb{E}[P(y=1 \mid x_t, \theta) \mid \text{data}]$ from the flat Bayesian logit (Sec. VI); that is still one probability per day—for example long when $\bar{p}_t \geq h$ for user-chosen h —and does not, without further code, fold independent posterior draws into position sizing or transaction-cost-aware execution. With `STRATEGY_TUNE_HURDLE=1` (default), an inner chronological validation slice of pre-holdout rows picks a hurdle h^* by *direction accuracy*, then PyMC refits on *all* pre-holdout rows before scoring the terminal tail (two NUTS passes); disable with `STRATEGY_TUNE_HURDLE=0` to keep only the fixed `STRATEGY_HURDLE` row.

Practical interpretation of the proxy. On our May 2026 replay (Adam LR= 10^{-6} , 50 epochs, channels $\{128, 64\}$, MPS; `STRATEGY_COST_BPS=0`), buy-and-hold lost $\approx 47\%$ on this tail while several long-flat rules *reduced* drawdowns by sitting out part of the decline; the calibrated-threshold rule finished near flat in total return with materially lower volatility than full exposure (Table V). That is a **relative** risk management outcome on one bearish window—not a certificate of alpha, and not robust to costs, rolls, or seed/hardware drift.

TABLE V

Terminal holdout *proxy* economics: TCN_PCA_WHEAT_ONLY $\hat{p}_t$ with buy-and-hold vs. long-flat rules (958 days; frictionless simple returns; turnover cost = 0; no margin, rolls, or commissions; May 2026 MPS replay).

Rule	Tot. ret.	Ann. vol.	Sharpe	Max DD	% long
Buy-and-hold	-0.468	0.319	-0.36	-0.613	1.00
Long-flat ($\hat{p} \geq 0.52$)	0.000	0.000	—	0.000	0.00
Long-flat ($\hat{p} \geq 0.50$)	0.009	0.005	0.51	0.000	0.001
Long-flat ($\hat{p} \geq \tau$)	0.000	0.217	0.11	-0.400	0.42
Momentum (y_{t-1})	-0.155	0.211	-0.11	-0.399	0.46

Regenerate via Appendix H (script `scripts/wheat_strategy_holdout_validation.py`; match Table III TCN_* exports). Here τ is the TCNClassifier validation threshold (0.4849 in this run). Hurdle 0.52 can yield $w_t \equiv 0$ when $\hat{p}_t < 0.5$ throughout—a calibration artifact.

IX. DISCUSSION

What to take away. Predicting tomorrow’s *up vs. down* move in wheat from public daily data is genuinely difficult. Most of our cross-validation scores sit near coin-flip (≈ 0.5), so readers should treat any model as a weak hint, not a guarantee. The final holdout slice is just *one* future window; rankings can move if the calendar cut-off changes.

How the pieces fit. The deep learning block (TCN_PCA_WHEAT_ONLY) is our headline direction model on the story we told in the tables. The Bayesian logistic section (§VI) emphasizes *uncertainty* on the flattened wheat-PCA design—useful for probability-style dashboards, not a claim that the linear head beats sequence models on raw inputs. The long-flat table (§VIII) compares **buy-and-hold** to **long-flat** rules on frictionless proxy returns only; it does not price commissions, margin, options, rolls, or other execution detail. It is **not** investment advice and not a full trading simulation; a later term targets model finetuning plus closer-to-live transaction modeling.

Next steps. Concrete extensions (costs, more out-of-sample checks, tighter data hygiene) stay under **Future Work** below.

X. CONCLUSION

The central message is *forecast realism under honest evaluation*, not a “which architecture wins” leaderboard. Across four heterogeneous models on identical leakage-aware data, the TCN classifier posted the best *direction* metrics on our frozen holdout, yet cross-validation still sits near coin-flip—so the honest read is a *small* nonlinear lift over transparent baselines, not a proof of easy economic alpha. Accordingly, reported differences between models should be interpreted descriptively rather than as statistically significant superiority claims. Linear and random-feature models therefore remain valuable references: interpretable, stable, and close enough in accuracy that deployment would hinge on costs, monitoring, and how probabilities are used, not on a razor-thin accuracy gap

alone. Repository code also documents EMD cautions for signed targets and optional PCA-augmented TCN tracks (Sec. V) as *ablations*, not as extra headline champions. The Bayesian panel (§VI) and the long-flat proxy table (§VIII, Table V) reinforce the same point from different angles: when the directional signal is thin, *how* you use probabilities—calibration, uncertainty, when to stay flat—can read larger than a point estimate of accuracy; on our single bearish holdout tail, avoiding full exposure mattered more for a toy P&L story than bragging rights on accuracy alone.

Future Work. Reasonable near-term refinements—without committing to a full institutional production stack—include the following.

- (a) *Daily execution checklist (minimal P&L read-out).* `wheat_strategy_holdout_validation.py` now supplies the zero-cost prototype in Sec. VIII; extend it with explicit round-trip commissions and roll basis matching a defended brokerage template, inner-validation-only tuning of h (never on the terminal fifteen percent), and stress tests over alternate calendar cutoffs.
- (b) *Extra out-of-sample checks.* Split the timeline into an additional train/validation block or repeat the hold-out exercise with alternate calendar cutoffs to see whether rankings among the four models stay stable rather than overstating conclusions from one frozen tail slice. A follow-on walk-forward design should use expanding origins, nested model selection, block-bootstrap AUC intervals, FRED-vintage sensitivity checks, and broker-realistic cost metrics.
- (c) *Light-touch uncertainty use.* Retain the Bayesian logistic path (§VI) but visualize a few posterior uncertainty bands for class probabilities or compare one calibration-aware threshold to the default 0.5 cut, stopping short of portfolio optimization or latent regime estimation.
- (d) *Targeted feature ablations.* On the existing pipeline, document one or two clean comparisons—for example TCN with versus without the macro PCA block on identical splits—before expanding to new raw datasets; any added series should stay within public sources with clearly documented lags.
- (e) *Broker-realistic strategy simulation (planned follow-on).* Beyond the frictionless buy-and-hold vs. long-flat prototype in Sec. VIII, extend the execution layer with parameterized commissions and rolls, margin and contract-multiplier accounting, and (where relevant) gap and liquidity assumptions, coupled with continued model finetuning, so reported economics approach what a disciplined wheat-futures backtest would require.

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APPENDIX A: SELECTED FRED-MD VARIABLES AND T-CODES

Table VI lists the 31 macro features and integer t-codes used by the codebase (1 level, 2 first difference, 5 log first difference, 6 log second difference).

APPENDIX B: EXTENDED LIMITATIONS

Microstructure and calendar effects. Daily closes may not be perfectly comparable across daylight-saving, holiday-shortened weeks, or contract roll assumptions if continuous series were stitched.

Parameter stability. TCN early stopping and threshold rules depend on internal validation slices; small changes can alter reported accuracy by a percentage point.

Macro stale-ness. Forward-filled monthly indicators evolve slowly; much apparent “macro content” may be

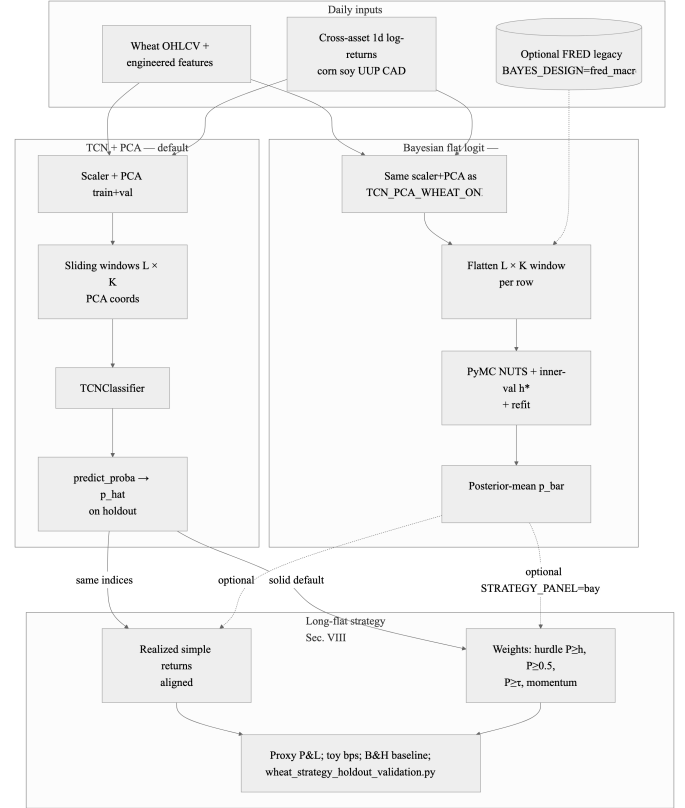


Fig. 8. Repository pipeline: Sec. VIII long-flat proxy (Table V) uses **TCN predict_proba** $\hat{p}_t$ by default; **STRATEGY_PANEL=bayes** adds the same rules on **posterior-mean** $\bar{p}_t$ from the wheat-PCA Bayesian logit (Sec. VI), including optional inner-validation hurdle h^* (two NUTS fits). Legacy **BAYES_DESIGN=fred_macro** remains for appendix-style macro+lag designs and older figure exports.

collinear with smoothed price trends already encoded in lags.

Price level versus return metrics. Commodity work sometimes reports high R^2 on price levels built from lags plus predicted changes; such fit often reflects persistence rather than economically sharp knowledge of Δ_t , so reporting quantiles or densities of Δ_t can be more informative than a single thresholded sign.

Alternative hypotheses. Formal tests for equality of AUCs across models (e.g., DeLong) were not run; claims are descriptive.

APPENDIX C: METRIC GLOSSARY FOR DIRECTION TASKS

Accuracy is the fraction of days where $\hat{y}_t = y_t$. Random guessing with matched marginal frequencies typically lies near 0.5 for balanced designs but can differ under skewed priors.

Macro-averaged F1 averages the F1 scores of the positive and negative classes, penalizing models that collapse to always predicting a single class even if overall accuracy appears acceptable.

ROC-AUC measures the probability that a randomly sampled up day receives a higher predicted score than a randomly sampled down day; unlike accuracy, it is insensitive to a fixed threshold but still assumes i.i.d. pairs—a mild abuse under serial correlation unless blockwise comparisons are used.

RMSE on Δ gauges economic magnitude of price-change errors in dollars per bushel per day for the regression heads. Values near 15 USD with daily prices near hundreds imply non-trivial relative errors even when direction is occasionally inferred correctly.

Confusion matrix cells allow inspection of asymmetric costs: false positives on up calls when leverage is long-

biased may be more costly than false negatives depending on a user’s utility function.

APPENDIX D: REPLICATION HYGIENE CHECKLIST

Direction forecasts on stitched wheat and FRED-MD panels are easy to mis-specify: labels, lags, and transforms must stay aligned with calendar dates, and any information leak across the chronological split inflates metrics silently. The list below is a practical guardrail for anyone re-running the repository scripts or extending them—covering data layout, fold-safe preprocessing, environment provenance, and documentation habits that make reported numbers defensible.

- 1) Confirm wheat CSV and both FRED vintage directories exist under `data/` before launching scripts.
- 2) Never call `fit_transform` on the concatenated train+test panel; always fit inside each training fold.
- 3) Record `git` commit hash and library versions (`torch`, `sklearn`) in lab notes.
- 4) For TCN, log whether CUDA, MPS, or CPU backends were active.
- 5) Archive plots immediately after runs; filenames in this PDF assume default script outputs.
- 6) When comparing to external papers, align definitions of up/down (close-to-close vs. close-to-open).
- 7) Re-run with CPU-only seeds if bitwise reproducibility is required.
- 8) Document any manual edits to CSV rows or missing macro fills.
- 9) For PCA-augmented pipelines, store explained variance ratios per fold.
- 10) Verify that classification labels match regression-implied signs on a handful of dates each release.

APPENDIX E: MACRO TICKER INDEX

The codebase macro block is identical to Table VI: 31 FRED-MD symbols with the integer t -codes and grouping shown there (`SELECTED_FEATURES` in Python).

Companion notes: dated runs, environment fingerprints, and supplementary metric tables for the exploratory Bayesian logistic run are archived as `bayes_mcmc_wheat_run_log.md`.

APPENDIX F: EXTENDED NARRATIVE ON WHEAT PRICE DISCOVERY

The Chicago Board of Trade wheat contract aggregates dispersed private information about planting progress, crop conditions, and export demand. Directional forecasts

TABLE VI
FRED-MD variables in `SELECTED_FEATURES`.

Symbol	Description (abbrev.)	t-code
RPI, W875RX1, CMRMTSPLx, IPFPNSS	Real activity & IP	5
USWTRADE, USTRADEx	Trade sales	5
BUSLOANS, CONSPI	Credit & reserves	5
S&P 500, S&P PE ratio	Equity levels	5
FEDFUNDS, TB3MS, TB6MS, GS1, GS5, GS10, AAA, BAA	Policy & yields	2
TB3SMFFM, TB6SMFFM, T1YFFM, T5YFFM, T10YFFM	Curve spreads vs. funds	1
AAAFFM, BAAFFM	Credit spreads vs. funds	1
EXSZUSx, EXJPUSx, EXUSUKx, EXCAUSx	FX vs. USD	5
PPICMM	Metals PPI component	6
UMCSENTx	Michigan sentiment	2

therefore compete not only with a martingale null but with highly informed commercial hedgers who observe localized weather and basis conditions unavailable in national macro releases.

Our macro panel proxies global financial conditions rather than field-level shocks; accordingly, marginal predictive power for *daily* signs is expected to be small. The modest TCN uplift is consistent with nonlinear interactions between risk sentiment (equities), the dollar, and interest rates—channels that linear ARX models approximate only crudely.

From a risk-management perspective, even a few percentage points of directional accuracy could matter if translated into options strategies with convex payoffs, but estimation error in AUC of order $1/\sqrt{n_{\text{test}}}$ for $n_{\text{test}}=955$ is itself several percentage points, so overlap of confidence sets across models is likely.

Future datasets might incorporate satellite greenness indices, soil moisture reanalysis, and oceanic ENSO indicators at sub-monthly resolution; such features would better align predictors with the physical drivers of yield surprises. Doing so would also reduce reliance on slow-moving macro factors that chiefly matter through financial channels.

Finally, reproducible pipelines should archive *exact* CSV hashes and random seeds; the present study recorded software versions alongside the repository and stored pickled models under `models/` for auditors to reload without retraining.

APPENDIX G: BAYESIAN MCMC SUPPLEMENTARY MATERIAL

Replication. From the repository root, executing `BAYES_DRAWS=10000 BAYES_TUNE=10000 ./venv/bin/python bayes_logit_mcmc_macro_pca.py` recreates the Table IV headline within NUTS noise; shorter 2000/6000 runs match at displayed precision (Table VII; `bayes_mcmc_wheat_run_log.md`). A convenient default is `BAYES_DRAWS=BAYES_TUNE=2000`. Appendix H lists this script alongside the other forecasting pipelines.

Why we made Table VII. PyMC lets us spend more CPU on tuning and draws, but we wanted a concrete check that extra samples were not silently changing the headline story. So we logged three budgets on the same script path and compared wall time, ESS, and the same sklearn metrics we already use elsewhere.

Low-level read of the three rows.

- **Wall time.** About 17s $\rightarrow$ 52s $\rightarrow$ 93s as we moved from 2000+2000 to 6000+6000 to 10000+10000

tune+draw settings (Apple Silicon, project `venv`; Yahoo cross-asset download excluded).

- **Minimum bulk ESS.** Worst marginal among α , σ_β , and each z_j rises from $\sim 2.5 \times 10^3$ to $\sim 8.7 \times 10^3$ to $\sim 2.0 \times 10^4$, matching longer chains.
- **Hold-out test (posterior mean).** Accuracy at 0.5 stays ≈ 0.542 ; ROC-AUC stays ≈ 0.508 at three decimals; Brier stays ≈ 0.249 .
- **Train block.** Posterior-mean train accuracy ≈ 0.520 and Brier ≈ 0.249 barely move—the bottleneck is shallow linear signal, not Monte Carlo noise once $\hat{R} \approx 1$.

What we take away. We did *not* see 10+10k draws unlock better holdout AUC or Brier at this print precision. For coursework, 2000+2000 already sits in the flat part of the curve; we lengthen chains only for prettier ESS. Independent-trial `sklearn` accuracies near 0.50 at threshold 0.5 contrast with ≈ 0.542 for the shrunk posterior mean, even when both AUC values agree within ~ 0.001 ; compare Brier and ROC, not accuracy alone.

What “posterior mean” meant in our code (plain English). For each retained draw $(\alpha^{(s)}, \beta^{(s)})$ with $\beta_j^{(s)} = \sigma_\beta^{(s)} z_j^{(s)}$, we evaluate $\text{logit}^{-1}(\alpha^{(s)} + \mathbf{x}^\top \beta^{(s)})$ and average those predictive probabilities across draws—*not*, in general, logit^{-1} applied to posterior mean coefficients. We scored the resulting $\bar{p}_t$ with `sklearn`’s `accuracy_score` (threshold 0.5), ROC-AUC, and Brier. We did **not** implement a different pipeline where each day draws a brand-new β from the chain and then majority-votes labels; Section VI ties the math to the actual script.

Small bookkeeping items we had to keep straight. Wall-clock rows exclude CSV loading. The holdout row count reads 958 for default `wheat_pca`; archived `fred_macro` tables used 956 rows versus 955 sequence rows elsewhere—we do not treat one-observation gaps as a model win. Minimum ESS in the table is the worst row in ArviZ among α , σ_β , and every z_j (legacy `fred_macro`: intercept plus ten β_j only).

ArviZ labels (only for the old ten-slope macro run). When we inspected the legacy `fred_macro` output, `beta[0]–beta[4]` lined up with the short-horizon return block and `beta[5]–beta[9]` with the macro PCs. For `wheat_pca`, the vector is longer; we trust the script’s column order instead of memorizing indices.

What we did not pursue here. We did not build prior-predictive dashboards or utility-based position sizing; those are natural extensions if we continue the project beyond this report.

TABLE VII

Three NUTS budgets on **default wheat_pca** (tune+draw rows): wall time, worst-case bulk ESS across $\alpha, \sigma_\beta, z_{1:p}$ in ArviZ, and holdout / train scores from *posterior-mean* probabilities.

Tune + draw	Wall clock	Min. ESS _{bulk}	Hold-out test			Train in-sample	
			Acc _{test}	ROC-AUC _{test}	Brier _{test}	Acc _{train}	Brier _{train}
2000 + 2000	≈17 s	≈2.5×10 ³	0.542	0.508	0.249	0.520	0.249
6000 + 6000	≈52 s	≈8.7×10 ³	0.542	0.508	0.249	0.520	0.249
10000 + 10000	≈93 s	≈2.0×10 ⁴	0.542	0.508	0.249	0.520	0.249

APPENDIX H: REPRODUCIBILITY COMMANDS

How to validate Table III. Install dependencies (`requirements-handin.txt`), download wheat + FRED-MD into `data/`, activate the project `venv`, `cd` into the *package directory* that contains `ARX/`, `TCN/`, and `bayes_logit_mcmc_macro_pca.py` while keeping `data/` one level above the Python file that `pathlib` resolves (see README layout), then run the shell block below. Optional `TCN_HOLDOUT_FRAC=0` reproduces CV-only mode.

Shell replay fragment (paths relative to that package directory).

```
./venv/bin/python ARX/ARX_Revised.py --mode both
./venv/bin/python ELM/ELM_Revised.py --mode both
./venv/bin/python FAVAR/FAVAR_Revised.py --mode both
./venv/bin/python TCN/TCN_Revised.py --mode both
# PCA-TCN table: match hyperparameters printed in the caption
export TCN_LR=1e-6 TCN_EPOCHS=50 TCN_CHANNELS=128,64
./venv/bin/python TCN/TCN_PCA.py
./venv/bin/python TCN/TCN_PCA_WHEAT_ONLY.py
./venv/bin/python bayes_logit_mcmc_macro_pca.py
# Strategy table: preliminary long-flat / proxy P and L
# using TCN_PCA_WHEAT_ONLY holdout probabilities
export WHEAT_CSV=./data/wheat-futures/wheat_futures_daily.csv
export TCN_LR=1e-6 TCN_EPOCHS=50 TCN_CHANNELS=128,64
./venv/bin/python scripts/wheat_strategy_holdout_validation.py
# Optional posterior-mean Bayes long-flat panel:
# uses wheat PCA by default and requires PyMC plus ArviZ.
# STRATEGY_PANEL=bayes STRATEGY_BAYES_DRAWS=400 \
#   STRATEGY_BAYES_TUNE=400 \
#   ./venv/bin/python scripts/wheat_strategy_holdout_validation.py
# BAYES_DESIGN=fred_macro STRATEGY_PANEL=bayes ...   legacy FRED macro design
# STRATEGY_TUNE_HURDLE=0 ...                           skip inner-val h*
```

Verifier checklist. (1) Install `requirements-handin.txt` or equivalent; (2) download wheat + FRED-MD CSVs before training; (3) prefer the same toolchain (MPS/CPU) for apple-to-apples digits; (4) holdout summaries print after five-fold CV when `TCN_HOLDOUT_FRAC=0.15` (defaults in both PCA scripts).